

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of M.Sc.(Mathematics) Examination November 2019
MA 811 Mathematical Methods-I

Date: 06/11/2019, Wednesday **Time:** 01:30 p.m. to 02:15 p.m. **Maximum Marks:** 20

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. In this question paper, $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{Z}$ denotes the set of all integers. All notations are standard.

Q.1. Choose the correct answer for the following questions. [20]

1. If $L[f(t)](s)$ is the Laplace transform of $f(t)$, then $L[e^{2t} \cos 5t](s) = \underline{\hspace{2cm}}$.

(A) $\frac{5}{(s-2)^2+25}$

(C) $\frac{s-5}{(s-5)^2+4}$

(B) $\frac{s-2}{(s-2)^2+25}$

(D) $\frac{2}{(s-5)^2+4}$

2. If $L^{-1}[F(s)](t)$ is the inverse Laplace transform of $F(s)$, then $L^{-1}\left[\frac{e^{-\pi s}}{s+3}\right](t) = \underline{\hspace{2cm}}$, where $H_a(t)$ is the Heaviside function.

(A) $e^{-3(t-\pi)} H_\pi(t)$

(C) $e^{-\pi(t-3)} H_3(t)$

(B) $e^{3(t-\pi)} H_\pi(t)$

(D) $e^{\pi(t-3)} H_3(t)$

3. If $L[f(t)](s) = F(s)$ is the Laplace transform of $f(t)$, then $L[tf(t)](t) = \underline{\hspace{2cm}}$.

(A) $F'(s)$

(C) $F'(s) + F(s)$

(B) $-F'(s)$

(D) $sF'(s) + F(s)$

4. If $f : (0, \infty) \rightarrow \infty\mathbb{C}$ is a periodic function with period T , then Laplace transform of $f(t)$ is $\underline{\hspace{2cm}}$.

(A) $\frac{1}{1-e^{sT}} \int_0^T f(t) e^{-st} dt$

(C) $\frac{1}{e^{-sT}} \int_0^T f(t) e^{-st} dt$

(B) $\frac{1}{1-e^{-sT}} \int_0^T f(t) e^{-st} dt$

(D) $\frac{1}{e^{sT}} \int_0^T f(t) e^{-st} dt$

5. What is the Laplace transform of the Dirac-delta function $\delta_5(t)$?

(A) 1

(B) $\frac{1}{s-5}$

(C) e^{-5s}

(D) e^{5s}

6. The Wronskian of two functions $\sin x$ and $\cos(x - 1)$ is_____.

- Ⓐ 1 Ⓑ $-\cos 1$ Ⓒ $\sin 1$ Ⓓ $-\sin 1$.

7. Let V be an inner product space with inner product $\langle \cdot, \cdot \rangle$ and let $x, y \in V$. Then x and y are called orthogonal if _____.

- Ⓐ $\langle x, y \rangle = 0$ Ⓒ $\langle x, y \rangle \neq 1$
 Ⓑ $\langle x, y \rangle = 1$ Ⓓ $\langle x, y \rangle \neq 0$.

8. For Hermite polynomials $H_n(x)$, $\int_{-\infty}^{\infty} e^{-x^2} H_2(x) H_3(x) dx =$ _____.

- Ⓐ 1 Ⓑ $8\sqrt{\pi}$ Ⓒ $48\sqrt{\pi}$ Ⓓ 0

9. For Legendre polynomials $P_n(x)$, $P_1(77) =$ _____.

- Ⓐ 1 Ⓑ 7 Ⓒ 77 Ⓓ 78

10. If $Z[(a_n)](z)$ is the Z-transform of sequence (a_n) , then $Z[(-2)^n](z) =$ _____.

- Ⓐ $\frac{z}{z-2}$ Ⓑ $\frac{z}{z+2}$ Ⓒ $\frac{1}{z-2}$ Ⓓ $\frac{1}{z+2}$

11. The Fourier transform of the function $f(x) = \begin{cases} 1, & \text{for } -1 < x < 1, \\ 0, & \text{otherwise} \end{cases}$ is _____,
 $s \neq 0$.

- Ⓐ $\sqrt{\frac{2}{\pi}} \frac{\sin s}{s}$ Ⓑ $\sqrt{\frac{2}{\pi}} \frac{\cos s}{s}$ Ⓒ $\frac{\sin s}{s}$ Ⓓ $\frac{\cos s}{s}$

12. Let $f : [-\pi, \pi] \rightarrow \mathbb{C}$. Extend f as a 2π -periodic function on $\mathbb{R}$. If f satisfies Dirichlet conditions, then the Fourier series of f converges to _____, for every $x \in \mathbb{R}$, where $f(x^+)$ and $f(x^-)$ are the right hand limit and left hand limit of f at x .

- Ⓐ $\frac{f(x^+)}{2}$ Ⓒ $\frac{f(x^+) + f(x^-)}{2}$
 Ⓑ $\frac{f(x^-)}{2}$ Ⓓ $f(x^+) + f(x^-)$.

13. Which of the following function whose Fourier transform is itself?

- Ⓐ $e^{-\frac{x^2}{2}}$ Ⓒ e^{-x}
 Ⓑ $e^{-|x|}$ Ⓓ $\chi_{[-1,1]}(x)$.

14. If $F_c[f(x)](s)$, $F_s[f(x)](s)$ and $F[f(x)](s)$ are Fourier cosine transform, Fourier sine transform and Fourier transform of an odd integrable function $f(x)$ respectively, then_____.

- (A) $F[f(x)](s) = F_c[f(x)](s)$
 (C) $F[f(x)](s) = -iF_s[f(x)](s)$
 (B) $F[f(x)](s) = iF_c[f(x)](s)$
 (D) $F[f(x)](s) = F_s[f(x)](s)$

15. If f is twice differentiable function and $f''(x) \rightarrow 0$ as $|x| \rightarrow \infty$, then $F[f''](s) =$ _____.

- (A) $s^2 F[f](s)$
 (C) $is^2 F[f](s)$
 (B) $-s^2 F[f](s)$
 (D) $-is^2 F[f](s)$

16. If $F[f](s)$ is the Fourier transform of f , $f * g$ is the convolution product of f and g and $f, g \in L^1\mathbb{R}$, then $F[f * g](s) =$ _____.

- (A) $F[f](s) * F[g](s)$
 (C) $F[f](s)F[g](s)$
 (B) $F[f](s) + F[g](s)$
 (D) none of these

17. What is the period of $\cos\left(\frac{x}{3}\right)$?

- (A) π
 (B) 2π
 (C) 3π
 (D) 6π

18. For the given function $f(x) = |x|$, $-\pi \leq x \leq \pi$, the coefficient b_n of Fourier series with given function f is_____.

- (A) 0
 (B) $-\frac{4}{n^2\pi}$
 (C) $\frac{2(-1)^{n+1}}{n}$
 (D) $\frac{4(-1)^n}{n^2}$

19. Which of the following function is an even function of x ?

- (A) $x^3 \cos x$
 (C) $x \cos x - e^x$
 (B) $e^{x^2} x^5 \sin x$
 (D) $e^x - \cos x$

20. Let $f \in L^1[-\pi, \pi]$. Then the complex Fourier series of f is given by $\sum_{n=-\infty}^{\infty} c_n e^{inx}$, where $c_n =$ _____, $n \in \mathbb{Z}$.

- (A) $\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx$
 (C) $\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$
 (B) $\frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{inx} dx$
 (D) $\frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-inx} dx$

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Date: 06/11/2019, Wednesday **Time:** 02:15 p.m. to 04:30 p.m. **Maximum Marks:** 50

Instructions:

1. Section I and II must be attempted in separate answer sheets.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
 5. Figures to right indicate marks.
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SECTION-I

Q-2(a) Attempt any two of the following. [08]

(i) Find the Laplace transform of $\frac{1 - \cos 2t}{t}$ and $t^2 \sin t$.

(ii) Find the Fourier cosine series expansion of the function

$$f(x) = \begin{cases} \frac{1}{2} + x, & \text{for } -\frac{1}{2} < x \leq 0, \\ \frac{1}{2} - x, & \text{for } 0 < x < \frac{1}{2}. \end{cases}$$

(iii) Let $\mathbb{R}^3$ be the Euclidean inner product space. Use the Gram Schmidt process to transform the basis $x_1 = (1, 2, 2)$, $x_2 = (-1, 0, 2)$ and $x_3 = (0, 0, 1)$ into an orthonormal basis $\{u_1, u_2, u_3\}$.

(b) Attempt any six of the following. [12]

(i) Find the Fourier sine series for the function $f(x) = e^{5x}$, $0 < x < \pi$.

(ii) Solve the integral equation $\int_0^\infty f(x) \cos sx \, dx = \begin{cases} 1 - s, & \text{for } 0 \leq s \leq 1, \\ 0, & \text{for } s > 1. \end{cases}$

(iii) Using convolution theorem, find the inverse Laplace transform of $\frac{1}{s(s^2 + 1)}$.

(iv) For Hermite polynomials $H_n(x)$, show that

$$H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) = 0, \forall n \geq 1.$$

(v) State Rodrigue's formula for Jacobi polynomials $P_n^{(\alpha, \beta)}(x)$. Find $P_1^{(2, 3)}(20)$.

(vi) If $F_s[f](s)$ and $F_c[f](s)$ are Fourier sine transform and Fourier cosine transform of a function $f \in L^1[0, \infty)$ respectively, then show that

$$F_s[f(x) \sin ax](s) = \frac{1}{2}(F_c[f](s-a) - F_c[f](s+a)).$$

(vii) If $L[f(t)](s)$ is the Laplace transform of a periodic function f with period T , then show that $L[f](s) = \frac{1}{1-e^{-sT}} \int_0^T f(t)e^{-st} dt$.

(viii) Find the Z-transform of the sequence $(\sin(n\alpha))$, $\alpha \in \mathbb{R}$.

SECTION II

Q-3(a) Attempt any three of the following. [18]

(i) Using Laplace transform, solve the differential equation $y'' + 9y = \cos 2t$ subject to conditions $y(0) = y'(0) = 1$.

(ii) Find the inverse Laplace transform of $\frac{5s+3}{(s-1)(s^2+2s+5)}$.

(iii) Find the Fourier series for $f(x) = \begin{cases} -\pi, & \text{for } -\pi < x < 0, \\ x, & \text{for } 0 < x < \pi. \end{cases}$

(iv) Find the Fourier sine transform of e^{-kx} , $k > 0$. Evaluate the integral $\int_0^\infty \frac{x^2}{(x^2+4)(x^2+9)} dx$. State the result which you have used?

(v) Find the Green's function for the differential equation $y''(x) = x^2$ subject to conditions $y(0) = 0, y(1) = 0$ and hence solve it.

(b) Attempt any two of the following. [12]

(i) Using inversion integral method, find the inverse Z-transform of $\frac{2z}{(z-1)(z^2+1)}$.

(ii) Solve the partial differential equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ ($x \in \mathbb{R}, t > 0$) subject to conditions $u(x, 0) = f(x)$ and $\frac{\partial u}{\partial t}(x, 0) = 0, \forall x \in \mathbb{R}$ and both $u, \frac{\partial u}{\partial x} \rightarrow 0$ as $|x| \rightarrow \infty$.

(iii) Compute the half-range Fourier sine series of $f(x) = \pi x - x^2, 0 < x < \pi$. Use Parseval's identity to evaluate the sum of the series $\sum_{n=1}^\infty \frac{1}{(2n-1)^6}$.

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